

A DATA-DRIVEN SPATIALLY ADAPTIVE SPARSE GENERALIZED LINEAR MODEL FOR FUNCTIONAL MRI ANALYSIS

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ABSTRACT

A novel data-driven sparse generalized linear model (GLM) and statistical analysis method for fMRI is developed. Although independent component analysis (ICA) has been broadly applied to fMRI to separate spatially or temporally independent components, recent studies show that ICA does not guarantee independence of simultaneously occurred distinct activity patterns in the brain and sparsity of the signal has been shown to be more important. Motivated from the ICA and biological findings such as sparse coding in the primary visual cortex simple cells, we propose a compressed sensing based data-driven sparse GLM solely based upon the sparsity of the signal. It enables estimation of spatially adaptive design matrix from sparse signal components that represent synchronous neural hemodynamics. Furthermore, an MDL based model order selection rule can determine unknown sparsity for sparse dictionary learning.

Index Terms— Sparse GLM, sparse dictionary learning, K-SVD, data-driven fMRI analysis, compressed sensing

1. INTRODUCTION

Independent component analysis (ICA) has been broadly applied to data-driven fMRI analysis due to its capacity to isolate functional spatial patterns that contain spatially localized neural dynamics from the data mixed with other simultaneous time-varying effects [1]. It models the observed vector $\mathbf{y} = [y_1, y_2, \dots, y_m]^T$ as a mixture of the source vector $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$, as follows:

$$\mathbf{y} = \mathbf{A}\mathbf{x} \quad (1)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is a mixing matrix. The ICA aims to find an unmixing matrix $\mathbf{W} \in \mathbb{R}^{n \times m}$ such that output vector $\mathbf{s} = [s_1, s_2, \dots, s_n]^T = \mathbf{W}\mathbf{y}$ provides estimates of all n spatially or temporally independent source signals. The ‘HYBICA’ approach [2] using a spatial ICA (sICA) and the unified ‘SPM-ICA’ method [3] using a temporal ICA (tICA) were recently proposed to combine ICA with statistical parametric mapping (SPM) to construct the data-driven design matrix. Accordingly, many ICA algorithms such as Infomax and FastICA are being implemented for group analysis of fMRI such as a group ICA of fMRI Toolbox (GIFT) [4] or a tensor probabilistic ICA (PICA) [5].

However, recent studies [1, 6] showed that the most influential factor for ICA algorithm is the sparsity of the components rather than independence, and many hemodynamics are rarely independent from each other due to interconnections between biological networks. Furthermore, a set of receptive fields learned by maximizing sparseness in the output of a neural network model is shown to be spatially localized and selective to spatial structure at a specific scale, similar to cortical simple cells [7]. Similarly, electrophysiological experiments showed that a single neuron in the left posterior hippocampus was selectively fired by different views or drawings of a specific individual, while there was no statistically significant responses to other pictures or at the baseline level [8].

Therefore, we develop a novel *data-driven sparse* GLM framework for a maximum likelihood (ML) estimation of spatially adaptive design matrices solely based on the sparsity of underlying hemodynamic signals in fMRI. Here, the BOLD signal at a specific voxel may be regarded as a combination of a sparse set of dynamic components, where each component has different time-series signal patterns. Assuming that the components for each voxel are sparse and the neural integration of the dynamics is linear, applying the sparse dictionary learning algorithm would be reasonable to identify each component. Our framework automatically provides spatially adaptive design matrix. We also show that the unknown sparsity level can be estimated by minimum description length principle (MDL) [9]. Using simulation and the experiment results, we show that the proposed method is more individually adaptive and extracts the neural dynamics more accurately compared to the conventional methods.

2. THEORY

2.1. Data-driven Sparse GLM

Let $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_N] \in \mathbb{R}^{m \times N}$ and $\mathbf{E} = [\boldsymbol{\varepsilon}_1, \dots, \boldsymbol{\varepsilon}_N] \in \mathbb{R}^{m \times N}$, where $\mathbf{y}_i \in \mathbb{R}^m$ and $\boldsymbol{\varepsilon}_i \in \mathbb{R}^m$ represent samples of a BOLD signal and the corresponding noise at the i -th voxel, respectively. SPM assumes the following generalized linear model (GLM):

$$\mathbf{y}_i = \mathbf{D}\mathbf{x}_i + \boldsymbol{\varepsilon}_i, \quad i = 1, \dots, N, \quad (2)$$

where $\mathbf{D} \in \mathbb{R}^{m \times n}$ denotes the regressors, and $\mathbf{x}_i \in \mathbb{R}^n$ denotes the corresponding response signal strength at the i -

th voxel. Unlike the standard GLM that uses a pre-defined design matrix $\mathbf{D}$, the proposed model assumes $\mathbf{D}$ as an unknown *global* dictionary. In addition, the BOLD signal at each voxel is described as a sparse combination of atoms from $\mathbf{D}$ by assuming that the signal contribution is sparse (i.e. $\|\mathbf{x}_i\|_0 \leq k$). Then, assuming that noise at each voxel $\varepsilon_i \sim N(0, \sigma_i^2 \mathbf{I}_m)$ is independent of each other, the maximization of the log-likelihood under the sparsity constraint can be formulated by introducing unknown support set $\{I_i\}_{i=1}^N$ into a log-likelihood function:

$$l(\mathbf{D}, \mathbf{X}, \mathbf{A}, \{I_i\}_{i=1}^N) = -\frac{m}{2} \sum_{i=1}^N \log(2\pi\sigma_i^2) - \frac{1}{2} \sum_{i=1}^N \frac{(\mathbf{y}_i - \mathbf{D}_{I_i} \mathbf{x}_{I_i,i})^T (\mathbf{y}_i - \mathbf{D}_{I_i} \mathbf{x}_{I_i,i})}{\sigma_i^2} \quad (3)$$

where $\mathbf{x}_{I_i,i}$ denotes a subvector of $\mathbf{x}_i$ collected by elements in the index set I_i . Since maximizing Eq. (3) is a complicated non-convex nonlinear optimization problem, we address the problem by using alternative maximization. The unknown parameter $\mathbf{A}$ can be then calculated by differentiating Eq. (3) as:

$$\sigma_i^2 = \frac{1}{m} \mathbf{y}_i^T \mathbf{P}_{\mathbf{D}_{I_i}}^\perp \mathbf{y}_i \quad (4)$$

where $\mathbf{P}_{\mathbf{D}_{I_i}}^\perp$ denotes the projector associated with the orthogonal complement of the range space of $\mathbf{D}_{I_i}$. Next, if $\mathbf{A}$ is denoted as a diagonal matrix with i -th diagonal elements of σ_i and zeros elsewhere, we represent an equivalent optimization problem with respect to $\mathbf{D}$, $\tilde{\mathbf{X}}$:

$$\min_{\mathbf{D}, \tilde{\mathbf{X}}} \|\tilde{\mathbf{Y}} - \mathbf{D}\tilde{\mathbf{X}}\|_F, \quad \text{subject to } \|\mathbf{x}_i\|_0 \leq k, \quad (5)$$

where $\tilde{\mathbf{Y}} = \mathbf{Y}\mathbf{A}^{-\frac{1}{2}}$ and $\tilde{\mathbf{X}} = \mathbf{X}\mathbf{A}^{-\frac{1}{2}}$. Eq. (5) can be addressed by using a sparse dictionary learning algorithm such as the K-SVD [10]. Given a measurement $\mathbf{Y}$, the K-SVD algorithm search the best possible dictionary $\mathbf{D}$ for the sparse representation of $\mathbf{Y}$ using Eq. (5). It includes a two-step process per iteration: (i) sparse coding stage, and (ii) codebook update stage. More specifically, for a pre-defined dictionary estimate $\hat{\mathbf{D}}$, the sparse coding step solves the following for $i = 1, \dots, N$:

$$\min_{\mathbf{x}_i} \|\tilde{\mathbf{y}}_i - \hat{\mathbf{D}}\tilde{\mathbf{x}}_i\|_2^2, \quad \text{subject to } \|\tilde{\mathbf{x}}_i\|_0 \leq k. \quad (6)$$

We solve this by a simple thresholding based on correlation. The active index set I_i is estimated by collecting indices corresponding to the k -largest correlation between $\mathbf{y}_i$ and $\mathbf{d}_j$ as:

$$C_{\mathbf{y}_i}(j) = \frac{\|\mathbf{y}_i^T \mathbf{d}_j\|_2^2}{\sigma_i^2 \|\mathbf{d}_j\|_2^2}, \quad j = 1, \dots, n, \quad (7)$$

and the signal estimate $\hat{\mathbf{x}}_{I_i,i}$ is then given by

$$\hat{\mathbf{x}}_{I_i,i} = \left(\mathbf{D}_{I_i}^T \mathbf{D}_{I_i} \right)^{-1} \mathbf{D}_{I_i}^T \mathbf{y}_i. \quad (8)$$

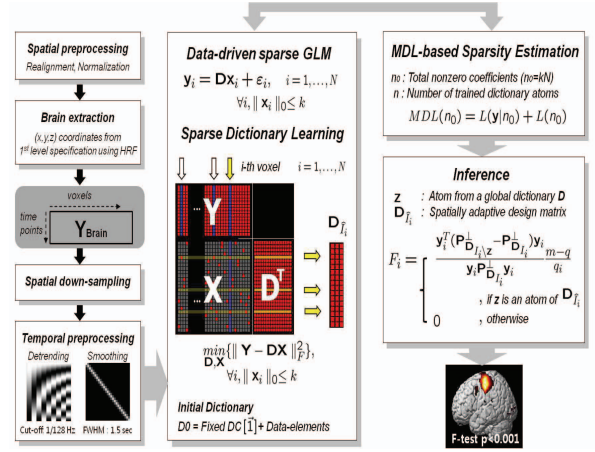


Fig. 1. Block diagram of the proposed method.

Then, in codebook update stage, K-SVD changes the columns of $\mathbf{D}$ sequentially with the corresponding coefficient row vector ($\mathbf{x}^j$). This can be solved using singular value decomposition (SVD) with sparsity constraint as following: (i) define the index set ω^p corresponding non-zero indices of $\tilde{\mathbf{x}}^p$, (ii) compute $\mathbf{E}_p = \tilde{\mathbf{Y}} - \sum_{j \neq p} \mathbf{d}_j \tilde{\mathbf{x}}^j$, (iii) define $\mathbf{\Omega}_p$ as a diagonal matrix with ones on the indices corresponding ω^p and zeros elsewhere, (iv) choose a subset $\mathbf{E}_p^{\mathcal{R}} = \mathbf{E}_p \mathbf{\Omega}_p$, (v) take SVD to the restricted $\mathbf{E}_p^{\mathcal{R}}$, and (vi) update $\hat{\mathbf{d}}_p$ using the first left eigenvector and $\tilde{\mathbf{x}}_p^{\mathcal{R}}$ using the first right eigenvector weighted with corresponding singular value. Note that we fixed a constant dictionary atom $\mathbf{d}_1 = [1, \dots, 1]^T \in \mathbb{R}^m$ to account for DC-bias and drift and did not update it during the process. Advantages of using correlation based thresholding in sparse coding stage is: (i) computational simplicity, and (ii) it retains the original BOLD signals in selecting correlated atoms, so that resulting sparse coding stage works as PCA for the collected time series.

2.2. MDL Principle For Sparsity Level

It is known that K-SVD is sensitive to sparsity level. So, we employed MDL method [9] as a criteria for sparse dictionary learning to determine the optimal k having the best trade-off between fidelity and model complexity. According to MDL principle, the best choice of model order n_0 to explain the data $\mathbf{y}$ is the one which minimizes

$$\text{MDL}(n_0) = \mathbf{L}(\mathbf{y}|n_0) + \mathbf{L}(n_0), \quad (9)$$

where $\mathbf{L}(\mathbf{y}|n_0)$ is the goodness-of-fit of $\mathbf{y}$ when encoded with n_0 , $\mathbf{L}(n_0)$ is the code length in bits to encode the model itself. In our problem, k number of regressors are distinctly selected in each voxel, so the total model order becomes $n_0 = kN$. Using Eq. (4) into Eq. (3), the code length for the goodness-of-fit is

$$\mathbf{L}(\mathbf{y}|n_0) = \frac{m}{2} \sum_{i=1}^N \log_2 \left(\frac{2\pi}{m} \mathbf{y}_i^T \mathbf{P}_{\mathbf{D}_{I_i}}^\perp \mathbf{y}_i \right) \text{ (bits)}, \quad (10)$$

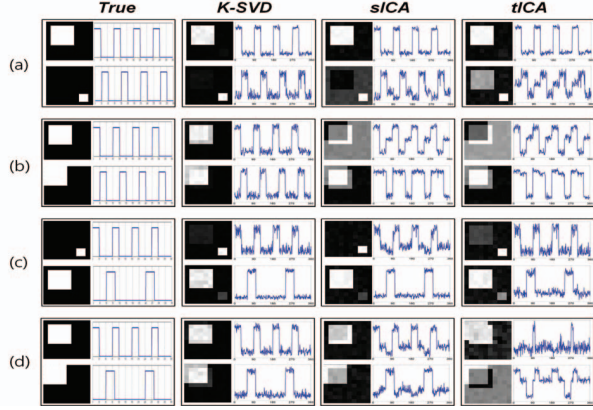


Fig. 2. Extracted time series and the corresponding z -maps for the four simulation scenarios are shown according to the following order: (a) temporally and spatially uncorrelated events, (b) spatially dependent events, (c) temporally dependent events and (d) temporally and spatially dependent events.

whereas, the code length $L(n_0)$ for MDL criterion can be described by the code length for the location of nonzero coefficient and magnitude as:

$$L(n_0) = \frac{1}{2}n_0 \log_2 n + n_0 \log_2 n = \frac{3}{2}n_0 \log_2 n, \quad (11)$$

where n is the number of dictionary atoms. Thus, we can solve the optimization problem Eq. (9) for various n_0 , and choose the n_0 that gives the minimum cost.

2.3. Activation Detection

We are interested in testing whether a specific temporal neural dynamics from a global dictionary $\mathbf{D}$ is present in a specific voxel. Hence, at each voxel, a binary hypothesis test is performed in which the null hypothesis $H_0 : \theta_i = 0$ is tested against the alternative hypothesis $H_1 : \theta_i \neq 0$, where θ_i is the response signal from

$$\mathbf{y}_i = \mathbf{z}\theta_i + \mathbf{D}_{I_i \setminus \mathbf{z}} \mathbf{x}_{I_i, i} + \varepsilon_i \quad (12)$$

where $\mathbf{z}$ denotes an atom from the global dictionary $\mathbf{D}$ that contains a neural dynamics of interest, and $\mathbf{D}_{I_i \setminus \mathbf{z}}$ denotes a reduced size local design matrix made by removing atom $\mathbf{z}$ from $\mathbf{D}_{I_i}$ (If $\mathbf{z}$ is not an atom of the local design matrix $\mathbf{D}_{I_i}$, then $\mathbf{D}_{I_i \setminus \mathbf{z}} = \mathbf{D}_{I_i}$). We can easily see when $\mathbf{z}$ does not belong to the set of atoms from $\mathbf{D}_{I_i}$, $\theta_i = 0$. Hence, H_0 holds. In other cases, we have

$$F_i = \frac{\mathbf{y}_i^T (\mathbf{P}_{\mathbf{D}_{I_i \setminus \mathbf{z}}}^\perp - \mathbf{P}_{\mathbf{D}_{I_i}}^\perp) \mathbf{y}_i}{\mathbf{y}_i^T \mathbf{P}_{\mathbf{D}_{I_i}}^\perp \mathbf{y}_i} \frac{m - q}{q_i} \quad (13)$$

where $\mathbf{P}_{\mathbf{D}_{I_i}}^\perp, \mathbf{P}_{\mathbf{D}_{I_i \setminus \mathbf{z}}}^\perp$ denotes the projection on the orthogonal complement on the range space of $\mathbf{D}_i$ and $\mathbf{D}_{I_i \setminus \mathbf{z}}$, respectively. The above process is illustrated in Figure 1.

3. METHOD

3.1. Simulation

We generated simulated data to test the validity of the proposed method in comparison to sICA, and tICA. Two pairs of 360 *sec* temporal waveforms were created and visual patterns comprised of box signals were repeated according to the four different scenarios as shown in Figure 2. Three kinds of visual patterns of 10×10 voxels were created, whose amplitudes were 1 in $\{2, \dots, 6\} \times \{2, \dots, 6\}$, $\{8, 9\} \times \{8, 9\}$, $\{1, \dots, 5\} \times \{1, \dots, 5\}$ and 0 elsewhere, respectively. Additionally, random white Gaussian noise with $\mathcal{N}(0, 0.11)$ was added. For the proposed method, we used 1-sparsity ($k = 1$) and the dictionary of size 3 ($n = 3$), which was learned using K-SVD algorithm with 20 iterations.

3.2. Real data analysis

We used the proposed method on the block-paradigm auditory stimulus task dataset of a single subject (SPM open-dataset) and event-related right finger tapping (RFT) task dataset of four subjects. The images were first spatially realigned, spatially normalized and resampled to $2 \times 2 \times 2$ (*mm*) voxels. Spatial smoothing with a $8 \times 8 \times 8$ (*mm*) FWHM Gaussian kernel was then applied only for the data to be used as a measurement in the sparse GLM model. After extracting the brain region, the BOLD signal was arranged as a m (time points) $\times N$ (voxels) matrix and was down-sampled 8 times along the spatial direction. Temporal detrending using a discrete cosine transform basis set with a cutoff frequency of $1/128$ Hz and temporal smoothing using 1.5 *sec* FWHM Gaussian kernel were followed. Then, the data were decomposed using the K-SVD, where correlation based thresholding method is used in sparse coding. An initial dictionary was determined by the data elements, and the first atom was fixed to be the DC component. A total of 30 iterations of K-SVD learning were performed for every dataset. The sparsity (k) was determined by MDL criterion by varying k from 1 to 10, and the number of dictionary atoms (n) was set to 40. After the dictionary learning, the nonzero k atoms were used as a local design matrix. Then, F -maps were calculated using Eq. (13) offline, and imported to SPM5 with degree of freedoms to obtain the activation map for a given p -value. We used an elementary contrast vector $\mathbf{c}$, i.e. $\mathbf{c} = \mathbf{e}_i \in \mathbb{R}^n$ that has zeros except 1 element at the i -th location, where the i -index varies depending on the neural dynamics of interest. It selects a regressor as $\mathbf{z} = \mathbf{D}\mathbf{c}$, so that we can test different atoms from a trained global dictionary $\mathbf{D}$.

4. EXPERIMENTAL RESULTS

In all the simulation scenarios, the time course extracted using the proposed method closely followed the original temporal responses, confirming that proposed method can effectively extract the sparse components. Moreover, the activation map by z -contrast follow the ground-truth accurately as shown in Figure 2. In Figure 3, the maximum number of sparsity (k) for the auditory stimulus dataset was chosen 2 according to our MDL criterion. The correlation coefficients

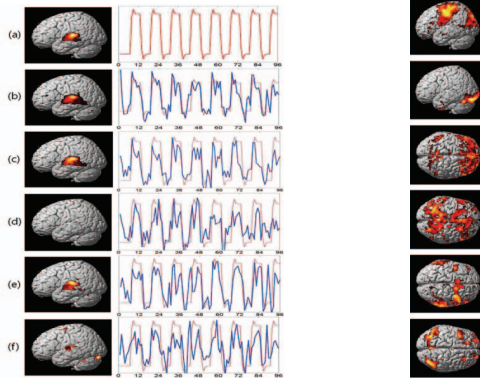


Fig. 3. F -maps for auditory stimulus dataset at a random field correction $p < 0.001$ using design matrices constructed by (a) a canonical HRF, (b) data-driven sparse GLM, (c) sICA (Infomax), (d) sICA (FastICA), (e) tICA (FastICA), and (f) PCA, respectively. On the right column, F -maps with respect to various non-task related signals are shown.

of each data-driven design matrix (blue lines) with canonical HRF (red lines) are: (b) 0.8214, (c) 0.8072, (d) 0.6045, (e) 0.7430, and (f) 0.5778, which showed that the extracted time course by the proposed algorithm are most correlated. On the other hand, MDL criterion selected the optimal sparsity level $k = 6, 6, 9$ and 7 for each subject in event-related RFT task dataset, respectively (see Figure 4). On the right sides of Figure 3 and Figure 4, the F -statistics maps corresponding other non-task related signal atoms by varying a contrast c . The maps indicate sparsely distributed patterns of task-related signals, nontask-related signals, motion artifacts and etc from distinct regions.

5. CONCLUSION

We proposed a new brain fMRI analysis using data-driven sparse GLM by exploiting sparsity of the components instead of independence, and it was shown that the unknown sparsity level can be estimated by MDL criterion. This new approach extracts individually adaptive activation patterns more accurately than conventional methods.

6. ACKNOWLEDGMENT

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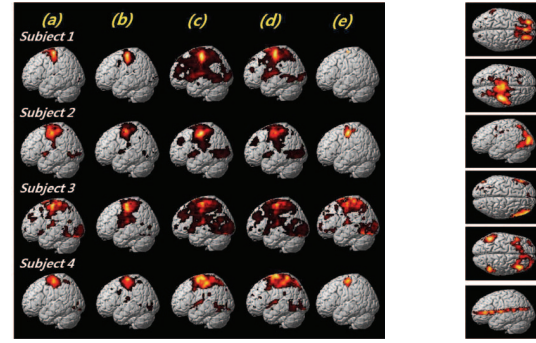


Fig. 4. F -maps for event-related RFT task dataset of 4 subjects at a random field correction $p < 0.001$ using the design matrices constructed by (a) canonical HRF, (b) data-driven sparse GLM, (c) sICA (Infomax), (d) sICA (FastICA), and (e) PCA. On the right column, F -maps with respect to various non-task related signals for subject 2 are shown.